color example code: colormaps_reference.py

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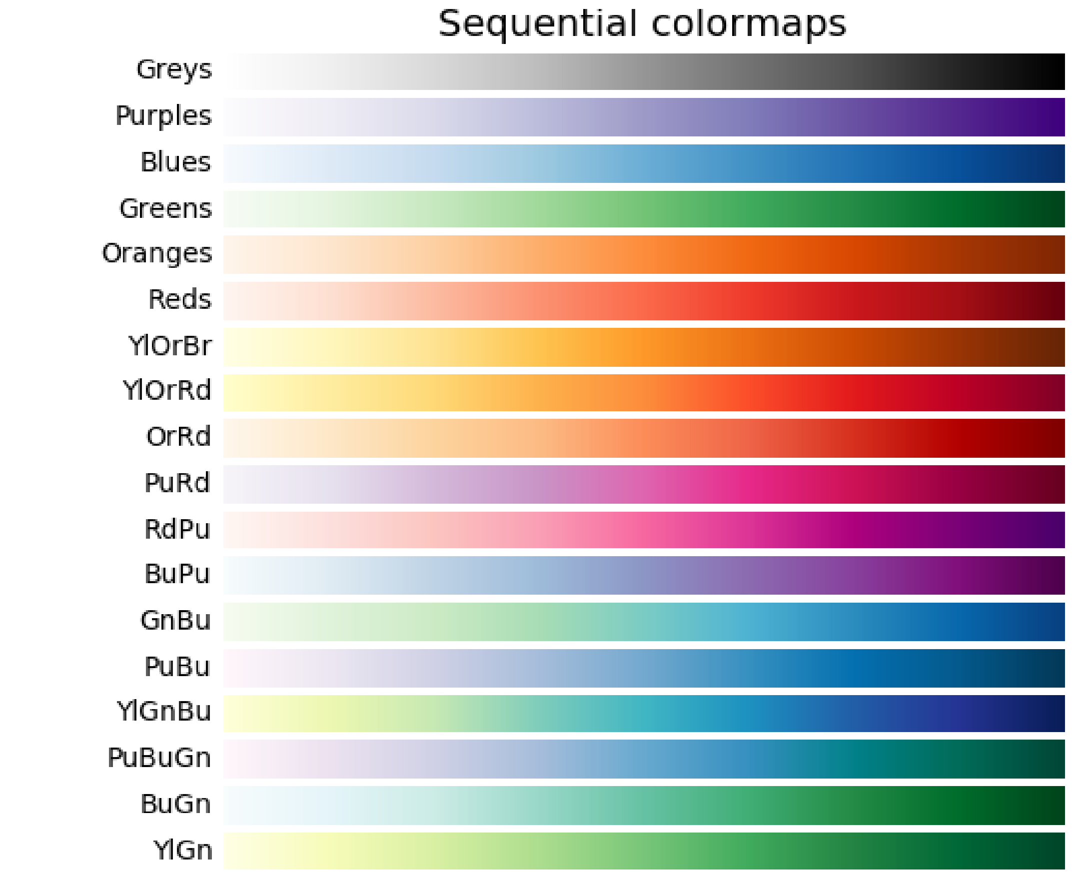

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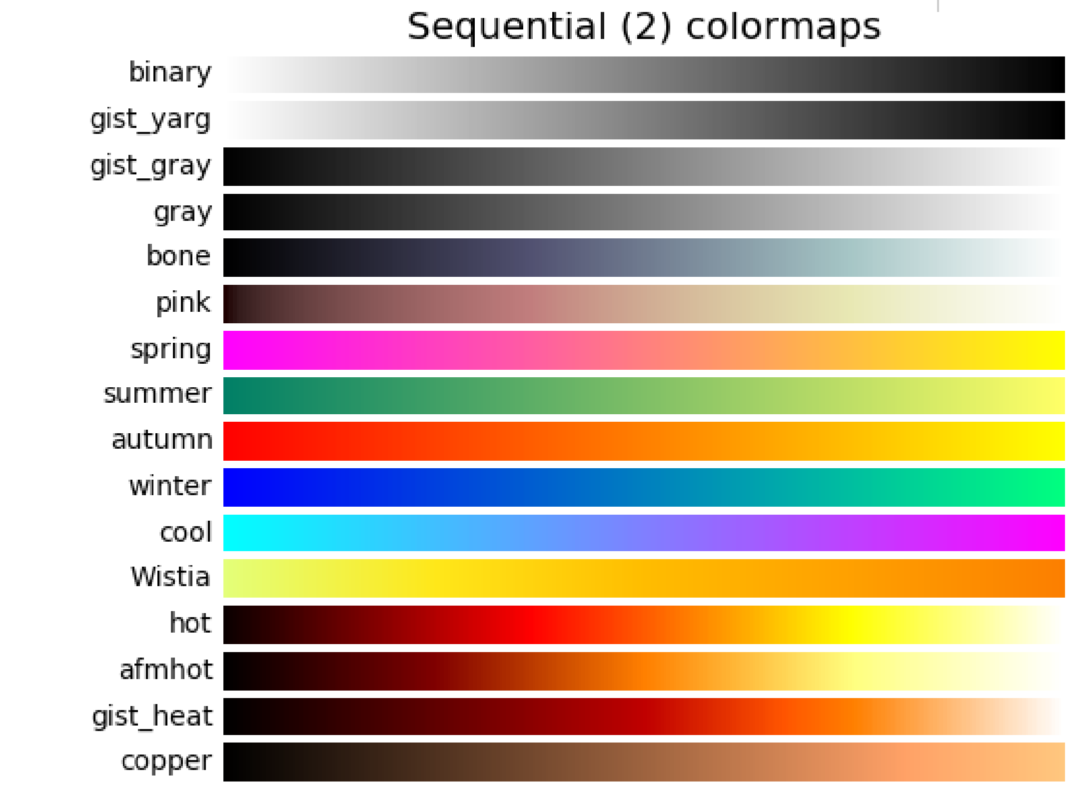

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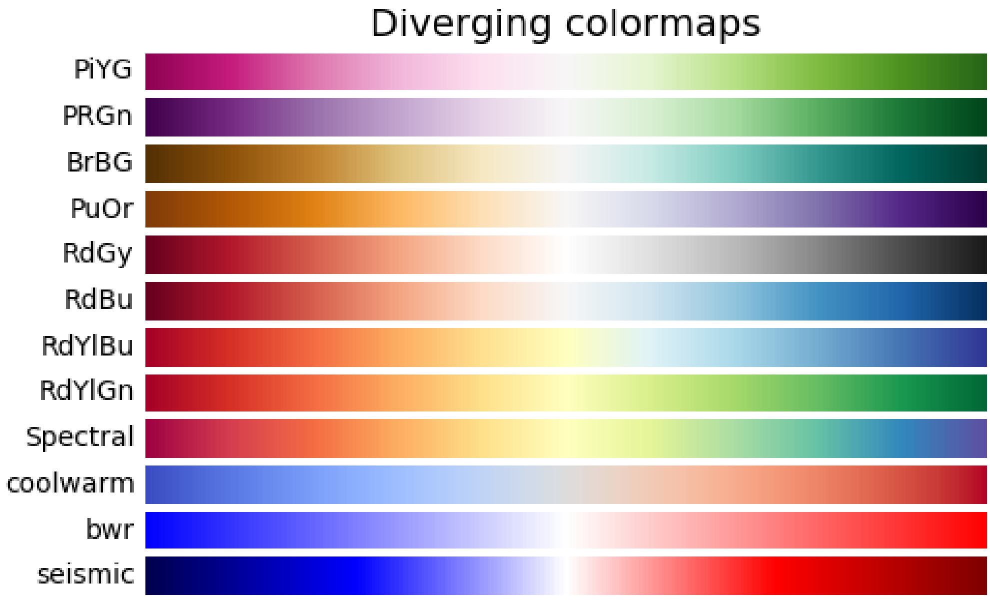

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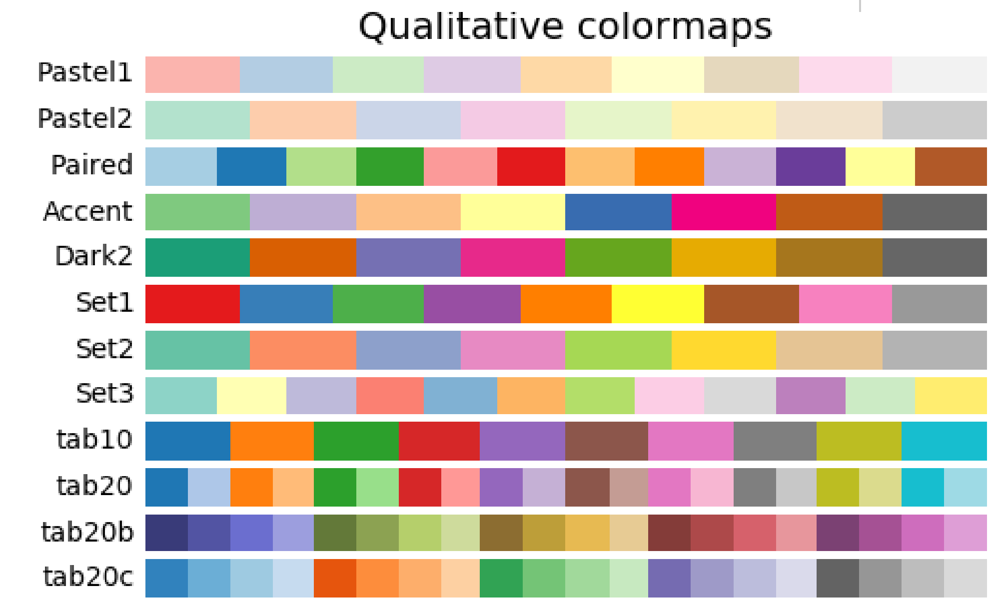

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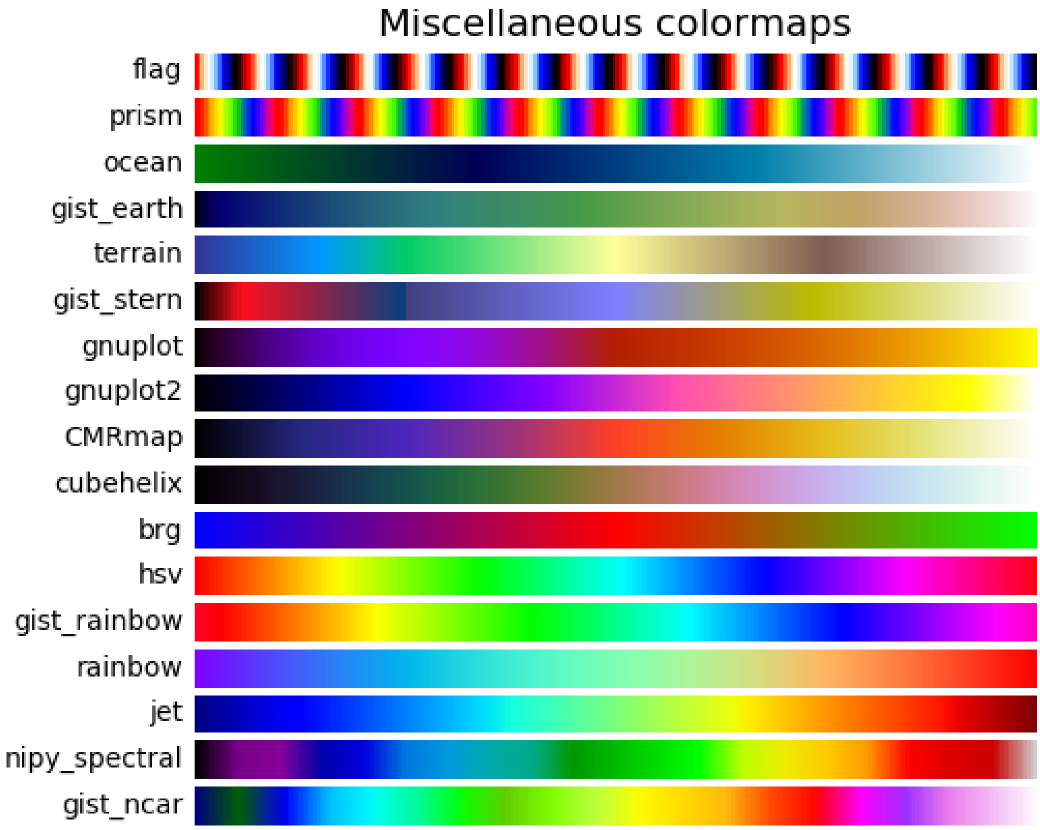

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```
"""
==================
Colormap reference
==================

Reference for colormaps included with Matplotlib.

This reference example shows all colormaps included with Matplo
any colormap listed here can be reversed by appending "_r" (e.g
These colormaps are divided into the following categories:

Sequential:
    These colormaps are approximately monochromatic colormaps v
    between two color tones---usually from low saturation (e.g.
    saturation (e.g. a bright blue). Sequential colormaps are i
    representing most scientific data since they show a clear p
    low-to-high values.

Diverging:
    These colormaps have a median value (usually light in color
    smoothly to two different color tones at high and low value
    colormaps are ideal when your data has a median value that
    (e.g.  0, such that positive and negative values are repres
    different colors of the colormap).

Qualitative:
    These colormaps vary rapidly in color. Qualitative colormap
    choosing a set of discrete colors. For example::

        color_list = plt.cm.Set3(np.linspace(0, 1, 12))

    gives a list of RGB colors that are good for plotting a ser
    a dark background.
```

```
Miscellaneous:
    Colormaps that don't fit into the categories above.

"""
import numpy as np
import matplotlib.pyplot as plt


# Have colormaps separated into categories:
# http://matplotlib.org/examples/color/colormaps_reference.html
cmaps = [('Perceptually Uniform Sequential', [
            'viridis', 'plasma', 'inferno', 'magma']),
         ('Sequential', [
            'Greys', 'Purples', 'Blues', 'Greens', 'Oranges', '
            'YlOrBr', 'YlOrRd', 'OrRd', 'PuRd', 'RdPu', 'BuPu',
            'GnBu', 'PuBu', 'YlGnBu', 'PuBuGn', 'BuGn', 'YlGn']
         ('Sequential (2)', [
            'binary', 'gist_yarg', 'gist_gray', 'gray', 'bone',
            'spring', 'summer', 'autumn', 'winter', 'cool', 'Wi
            'hot', 'afmhot', 'gist_heat', 'copper']),
         ('Diverging', [
            'PiYG', 'PRGn', 'BrBG', 'PuOr', 'RdGy', 'RdBu',
            'RdYlBu', 'RdYlGn', 'Spectral', 'coolwarm', 'bwr',
         ('Qualitative', [
            'Pastel1', 'Pastel2', 'Paired', 'Accent',
            'Dark2', 'Set1', 'Set2', 'Set3',
            'tab10', 'tab20', 'tab20b', 'tab20c']),
         ('Miscellaneous', [
            'flag', 'prism', 'ocean', 'gist_earth', 'terrain',
            'gnuplot', 'gnuplot2', 'CMRmap', 'cubehelix', 'brg'
            'gist_rainbow', 'rainbow', 'jet', 'nipy_spectral',


nrows = max(len(cmap_list) for cmap_category, cmap_list in cmap
gradient = np.linspace(0, 1, 256)
gradient = np.vstack((gradient, gradient))


def plot_color_gradients(cmap_category, cmap_list, nrows):
    fig, axes = plt.subplots(nrows=nrows)
    fig.subplots_adjust(top=0.95, bottom=0.01, left=0.2, right=
    axes[0].set_title(cmap_category + ' colormaps', fontsize=14

    for ax, name in zip(axes, cmap_list):
        ax.imshow(gradient, aspect='auto', cmap=plt.get_cmap(na
        pos = list(ax.get_position().bounds)
        x_text = pos[0] - 0.01
        y_text = pos[1] + pos[3]/2.
        fig.text(x_text, y_text, name, va='center', ha='right',

    # Turn off *all* ticks & spines, not just the ones with col
    for ax in axes:
        ax.set_axis_off()


for cmap_category, cmap_list in cmaps:
    plot_color_gradients(cmap_category, cmap_list, nrows)

plt.show()
```

Keywords: python, matplotlib, pylab, example, codex (see Search examples)